# Bachelor of Computer Applications

## Semester – V

## Business Analytics

## Experiment No.: 6

## Title: Sorting & Filtering Data


**Submitted By:**

Name: Prem Panchal

Reg. No.: 2411021240011


Date: 26/10/2026

GitHub Link: Prem2309/Business-Analytics-Projects

**Submitted To:**

Faculty Name: Mr. Nikhil Devetha


**Department of Computer Application**

**Alliance University**

Chandapura - Anekal Main Road, Anekal

Bengaluru – 562106

# Experiment No.: 6

# Sorting & Filtering Data

## 1. Introduction

This assignment demonstrates how sorting and filtering techniques can be used in Microsoft Power BI to examine sales transactions, identify high-value transactions, compare product performance, and locate underperforming products. The report presents transaction-level details alongside product-level revenue and profit summaries. A Region slicer is included to allow the report viewer to focus on selected regions and explore the sales information interactively.

## 2. Objectives

- Import and display the sales analysis dataset in Power BI.
- Organize transaction records so that individual sales can be inspected clearly.
- Use sorting to compare products and transactions by relevant sales measures.
- Use filtering to focus the report on selected regions.
- Compare product revenue and profit to identify strong and weak performers.
- Present the results in a readable report containing tables, a slicer, and a bar chart.
- Use the report to support business decisions about sales performance.

## 3. Dataset Overview

The dataset used for this experiment is “sales_analysis_dataset.xlsx”. It contains sales transaction information such as Transaction ID, Date, Region, Customer, Product, Quantity, Revenue and Profit. These fields allow the report to examine individual transactions and summarize performance by product. The report also uses product category information in the product summary table.

## 4. Report Design and Sorting/Filtering Process

### 4.1 Transaction Details Table

The transaction table displays Transaction ID, Date, Region, Customer, Product, Quantity, Revenue and Profit. This detailed view makes it possible to inspect sales records individually and compare the value of different transactions. The totals row summarizes the displayed quantity, revenue and profit.

### 4.2 Sorting Data

Sorting arranges records in ascending or descending order according to a selected field. In a sales report, sorting by Revenue helps bring higher-value transactions into view, while sorting by Profit helps compare the financial contribution of sales. Product-level summaries can also be sorted by revenue or profit to make differences between products easier to recognize. Sorting changes the display order; it does not remove records from the dataset.

### 4.3 Filtering Data with the Region Slicer

The report includes a Region slicer with East, North, South and West as selectable values. Selecting one or more regions filters the visuals to show the relevant sales information. This allows a manager to compare regional performance without changing the underlying dataset. Clearing the selection returns the report to the broader view.

### 4.4 Product Revenue and Profit Summary

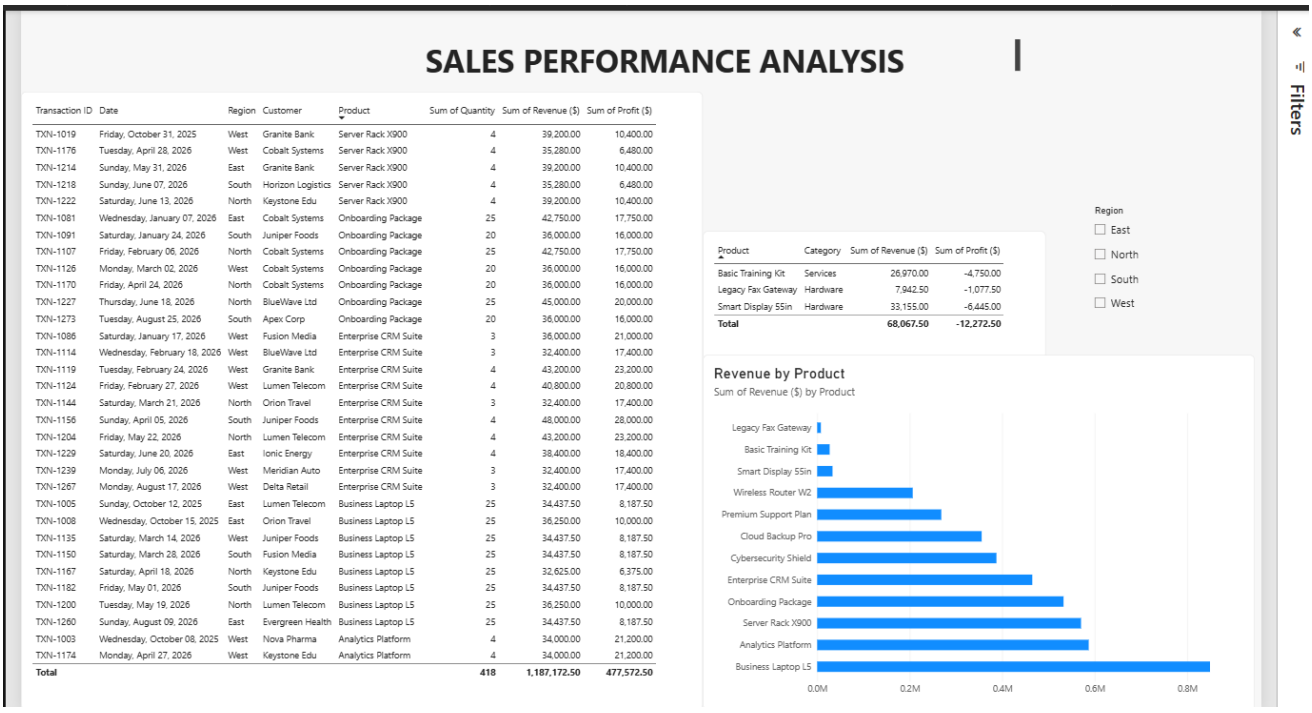
The product summary table compares product, category, revenue and profit. It highlights products with negative profit, including Basic Training Kit, Legacy Fax Gateway and Smart Display 55in in the displayed report. These products warrant further investigation because the shown profit values are below zero.

### 4.5 Revenue by Product Bar Chart

The horizontal bar chart compares revenue across products. Longer bars represent products with higher revenue. In the displayed report, Business Laptop L5 has the largest revenue bar, followed by Analytics Platform and Server Rack X900. The chart provides a quick visual ranking and complements the detailed table.

## 5. Final Power BI Report

The following screenshot shows the completed Sales Performance Analysis report. It combines a transaction table, a product-level summary, a Region slicer and a Revenue by Product chart in one report view.




| Transaction ID | Date | Region | Customer | Product | Sum of Quantity | Sum of Revenue ($) | Sum of Profit ($) |
|---|---|---|---|---|---|---|---|
| TXN-1019 | Friday, October 31, 2025 | West | Granite Bank | Server Rack X900 | 4 | 39,200.00 | 10,400.00 |
| TXN-1176 | Tuesday, April 28, 2026 | West | Cobalt Systems | Server Rack X900 | 4 | 35,280.00 | 6,480.00 |
| TXN-1214 | Sunday, May 31, 2026 | East | Granite Bank | Server Rack X900 | 4 | 39,200.00 | 10,400.00 |
| TXN-1218 | Sunday, June 07, 2026 | South | Horizon Logistics | Server Rack X900 | 4 | 35,280.00 | 6,480.00 |
| TXN-1222 | Saturday, June 13, 2026 | North | Keystone Edu | Server Rack X900 | 4 | 39,200.00 | 10,400.00 |
| TXN-1081 | Wednesday, January 07, 2026 | East | Cobalt Systems | Onboarding Package | 25 | 42,750.00 | 17,750.00 |
| TXN-1091 | Saturday, January 24, 2026 | South | Juniper Foods | Onboarding Package | 20 | 36,000.00 | 16,000.00 |
| TXN-1107 | Friday, February 06, 2026 | North | Cobalt Systems | Onboarding Package | 25 | 42,750.00 | 17,750.00 |
| TXN-1126 | Monday, March 02, 2026 | West | Cobalt Systems | Onboarding Package | 20 | 36,000.00 | 16,000.00 |
| TXN-1170 | Friday, April 24, 2026 | North | Cobalt Systems | Onboarding Package | 20 | 36,000.00 | 16,000.00 |
| TXN-1227 | Thursday, June 18, 2026 | North | BlueWave Ltd | Onboarding Package | 25 | 45,000.00 | 20,000.00 |
| TXN-1273 | Tuesday, August 25, 2026 | South | Apex Corp | Onboarding Package | 20 | 36,000.00 | 16,000.00 |
| TXN-1086 | Saturday, January 17, 2026 | West | Fusion Media | Enterprise CRM Suite | 3 | 36,000.00 | 21,000.00 |
| TXN-1114 | Wednesday, February 18, 2026 | West | BlueWave Ltd | Enterprise CRM Suite | 3 | 32,400.00 | 17,400.00 |
| TXN-1119 | Tuesday, February 24, 2026 | West | Granite Bank | Enterprise CRM Suite | 4 | 43,200.00 | 23,200.00 |
| TXN-1124 | Friday, February 27, 2026 | West | Lumen Telecom | Enterprise CRM Suite | 4 | 40,800.00 | 20,800.00 |
| TXN-1144 | Saturday, March 21, 2026 | North | Orion Travel | Enterprise CRM Suite | 3 | 32,400.00 | 17,400.00 |
| TXN-1156 | Sunday, April 05, 2026 | South | Juniper Foods | Enterprise CRM Suite | 4 | 48,000.00 | 28,000.00 |
| TXN-1204 | Friday, May 22, 2026 | North | Lumen Telecom | Enterprise CRM Suite | 4 | 43,200.00 | 23,200.00 |
| TXN-1229 | Saturday, June 20, 2026 | East | Ionic Energy | Enterprise CRM Suite | 4 | 38,400.00 | 18,400.00 |
| TXN-1239 | Monday, July 06, 2026 | West | Meridian Auto | Enterprise CRM Suite | 3 | 32,400.00 | 17,400.00 |
| TXN-1267 | Monday, August 17, 2026 | West | Delta Retail | Enterprise CRM Suite | 3 | 32,400.00 | 17,400.00 |
| TXN-1005 | Sunday, October 12, 2025 | East | Lumen Telecom | Business Laptop L5 | 25 | 34,437.50 | 8,187.50 |
| TXN-1008 | Wednesday, October 15, 2025 | East | Orion Travel | Business Laptop L5 | 25 | 36,250.00 | 10,000.00 |
| TXN-1135 | Saturday, March 14, 2026 | West | Juniper Foods | Business Laptop L5 | 25 | 34,437.50 | 8,187.50 |
| TXN-1150 | Saturday, March 28, 2026 | South | Fusion Media | Business Laptop L5 | 25 | 34,437.50 | 8,187.50 |
| TXN-1167 | Saturday, April 18, 2026 | North | Keystone Edu | Business Laptop L5 | 25 | 32,625.00 | 6,375.00 |
| TXN-1182 | Friday, May 01, 2026 | South | Juniper Foods | Business Laptop L5 | 25 | 34,437.50 | 8,187.50 |
| TXN-1200 | Tuesday, May 19, 2026 | North | Lumen Telecom | Business Laptop L5 | 25 | 36,250.00 | 10,000.00 |
| TXN-1260 | Sunday, August 09, 2026 | East | Evergreen Health | Business Laptop L5 | 25 | 34,437.50 | 8,187.50 |
| TXN-1003 | Wednesday, October 08, 2025 | West | Nova Pharma | Analytics Platform | 4 | 34,000.00 | 21,200.00 |
| TXN-1174 | Monday, April 27, 2026 | West | Keystone Edu | Analytics Platform | 4 | 34,000.00 | 21,200.00 |
| **Total** | | | | | **418** | **1,187,172.50** | **477,572.50** |

| Product | Category | Sum of Revenue ($) | Sum of Profit ($) |
|---|---|---|---|
| Basic Training Kit | Services | 26,970.00 | -4,750.00 |
| Legacy Fax Gateway | Hardware | 7,942.50 | -1,077.50 |
| Smart Display 55in | Hardware | 33,155.00 | -6,445.00 |
| **Total** | | **68,067.50** | **-12,272.50** |

*Figure 1: Sales Performance Analysis report showing transaction details, product summary, Region slicer and revenue chart.*

## 6. Key Observations

- The transaction table provides a detailed view of sales by transaction, date, region, customer and product.
- The totals displayed in the transaction table are Quantity: 418, Revenue: $1,187,172.50 and Profit: $477,572.50.
- Business Laptop L5 is the highest-revenue product in the displayed bar chart.
- Analytics Platform and Server Rack X900 also show comparatively high revenue in the chart.
- The product summary shows negative profit for Basic Training Kit (-$4,750.00), Legacy Fax Gateway (-$1,077.50) and Smart Display 55in (-$6,445.00).
- The Region slicer enables focused analysis of East, North, South and West.

## 7. Result

The sales report was organized to support sorting and filtering analysis. The transaction table presents individual sales records, the Region slicer allows interactive filtering, and the product summary and bar chart help compare revenue and profit. The report makes it easier to identify high-revenue products and products with negative profit.

| Area | Result |
|---|---|
| Transaction details | Displayed in a table with transaction, customer, product and financial fields |
| Sorting | Supports ordered comparison of transaction and product performance |
| Region filtering | East, North, South and West can be selected in the slicer |
| Highest revenue product | Business Laptop L5 in the displayed chart |
| Products with negative profit | Basic Training Kit, Legacy Fax Gateway and Smart Display 55in |
| Overall displayed totals | 418 quantity; $1,187,172.50 revenue; $477,572.50 profit |

## 8. Conclusion

Sorting and filtering are useful techniques for turning a large set of sales records into meaningful business information. In this Power BI report, the transaction table supports detailed inspection, the Region slicer makes regional filtering interactive, and the product table and bar chart reveal differences in revenue and profitability. The report indicates that Business Laptop L5 leads the displayed products by revenue, while several products show negative profit and may need additional review. These insights can help a sales manager prioritize further analysis and make more informed decisions.